Understanding Generative AI

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What is Generative AI?

Generative AI refers to artificial intelligence systems that can create new content rather than just analyzing existing data.

Traditional AI	Generative AI
Classifies emails as spam or not spam	Can write a completely new email for you

Three pillars that made it possible

Algorithms

Transformer architecture (2017) revolutionized processing of long text passages

Data explosion

Explosion of digital data (websites, code repositories, and other text) provided raw material for training these systems

Computing power

Massive increases in computational power (chips like GPUs) made it possible to train these models on all the data

How it works

Pre-training

Models analyze billions of text examples, learning to predict what comes next

Fine-tuning

Models are refined to follow instructions, be helpful, and avoid harmful content

Deployment

Users provide prompts, and the model generates responses based on the prompts and patterns it learned during training

Key capabilities

- Versatile language skills
- General-purpose abilities
- Learning from examples
- Connecting to external tools and data

Current limitations

- Knowledge cutoff date
- Potential inaccuracies ("hallucinations")
- Context window constraints
- Challenges with complex reasoning and math

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6 Techniques for Effective Prompt Engineering

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1. Provide context

Before

Tell me about climate change.

After

Explain three major impacts of climate change on agriculture in tropical regions, with examples from the past decade.

Be specific about what you want, providing details about scope, geography, timeframe, and other relevant parameters.

2. Show examples of what “good” looks like

Before

Please convert this technical statement to plain language: “The platform implements end-to-end encryption protocols to safeguard data integrity.”

After

Here are two examples of how to convert technical jargon into plain language:

1. “The quantum algorithm exhibits quadratic speedup.” → “The new method solves problems roughly twice as fast as previous methods.”
2. “The interface leverages intuitive design paradigms.” → “The design is easy to understand and use.”

Now, please convert this technical statement to plain language: “The platform implements end-to-end encryption protocols to safeguard data integrity.”

Providing examples helps the AI understand the pattern, style, or format you're looking for more clearly than descriptions alone.

3. Specify output constraints

Before

Design me a personal art portfolio website.

After

Create a clean, modern single-page portfolio website with these sections: Hero, About Me, Skills, Portfolio/Projects, Experience, and Contact. Make the navigation menu sticky and responsive, with hamburger menu on mobile. Use a sunset color palette and add a dark/light mode toggle.

Be specific about what you want, providing details about scope, geography, timeframe, and other relevant parameters.

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6 Techniques for Effective Prompt Engineering

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4. Break complex tasks into steps

Before

Analyze this quarterly sales data.

After

I'd like to analyze this quarterly sales data. Please approach this by:

1. Identifying top-performing products
2. Comparing current quarter to previous quarter
3. Highlighting unusual patterns or trends
4. Suggesting possible reasons for these trends

Breaking down complex tasks into clear steps guides the AI's reasoning process and ensures thorough, methodical responses.

5. Ask it to think first

Before answering, please think through this problem carefully. Consider the different factors involved, potential constraints, and various approaches before recommending the best solution.

Giving the AI space to think before responding encourages more thoughtful, comprehensive answers.

6. Define the AI's role

Please explain how rainbows form from the perspective of an experienced science teacher speaking to a bright 10-year-old who's interested in science.

Defining the AI's role, tone, or style helps shape its approach to fit your specific needs and audience.

Secret Weapon: Ask the AI for help with prompting

I'm trying to get you, Claude, to help me with [goal]. I'm not sure how to phrase my request to get the best results. Can you help me craft an effective prompt for this?

When you're not sure how to ask for something, the AI can help improve your prompt. This is perhaps the most powerful technique of all!

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AI Fluency: Key Terminology Cheat Sheet

Core AI Fluency Framework Terms

AI Fluency
The ability to work with AI systems in ways that are effective, efficient, ethical, and safe. Includes practical skills, knowledge, insights, and values that help you adapt to evolving AI technologies.

The 4Ds
The four core competencies of AI Fluency: Delegation, Description, Discernment, and Diligence.

Delegation
Deciding on what work should be done by humans, what work should be done by AI, and how to distribute tasks between them. Includes understanding your goals, AI capabilities, and making strategic choices about collaboration.

- **Problem Awareness**: Clearly understanding your goals and the nature of the work before involving AI.
- **Platform Awareness**: Understanding the capabilities and limitations of different AI systems.
- **Task Delegation**: Thoughtfully distributing work between humans and AI to leverage the strengths of each.

Description
Effectively communicating with AI systems. Includes clearly defining outputs, guiding AI processes, and specifying desired AI behaviors and interactions.

- **Product Description**: Defining what you want in terms of outputs, format, audience, and style
- **Process Description**: Defining how the AI approaches your request, such as providing step by step instructions for the AI to follow
- **Performance Description**: Defining the AI's behavior during your collaboration, such as whether it should be concise or detailed, challenging or supportive

Discernment
Thoughtfully and critically evaluating AI outputs, processes, behaviors and interactions. Includes assessing quality, accuracy, appropriateness, and determining areas for improvement.

- **Product Discernment**: Evaluating the quality of what AI produces (accuracy, appropriateness, coherence, relevance)
- **Process Discernment**: Evaluating how the AI arrived at its output, looking for logical errors, lapses in attention, or inappropriate reasoning steps
- **Performance Discernment**: Evaluating how the AI behaves during your interaction, considering whether its communication style is effective for your needs

Diligence
Using AI responsibly and ethically. Includes making thoughtful choices about AI systems and interactions, maintaining transparency, and taking accountability for AI-assisted work.

- **Creation Diligence**: Being thoughtful about which AI systems you use and how you interact with them
- **Transparency Diligence**: Being honest about AI's role in your work with everyone who needs to know
- **Deployment Diligence**: Taking responsibility for verifying and vouching for the outputs you use or share

Human-AI Interaction Modes

Automation
When AI performs specific tasks based on specific human instructions. The human defines what needs to be done, and the AI executes it.

Augmentation
When humans and AI collaborate as thinking partners to complete tasks together. Involves iterative back-and-forth where both contribute to the outcome.

Agency
When humans configure AI to work independently on their behalf, including interacting with other humans or AI. The human establishes the AI's knowledge and behavior patterns rather than specifying exact actions.

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AI Technical Concepts

Generative AI
AI systems that can create new content (text, images, code, etc.) rather than just analyzing existing data.

Large language models (LLMs)
Generative AI systems trained on vast amounts of text data to understand and generate human language.

Claude
Anthropic's family of large language models.

Parameters
The mathematical values within an AI model that determine how it processes information and relates different pieces of language to each other. Modern LLMs contain billions of parameters.

Neural networks
Computing systems similar to, but distinct from, biological brains. Composed of interconnected nodes organized in layers that learn patterns from data through training.

Transformer architecture
The breakthrough AI design from 2017 that enables LLMs to process sequences of text in parallel while paying attention to relationships between words across long passages.

Scaling laws
As AI models have grown larger and trained on more data with more computing power, their performance has improved in consistent patterns. This is an empirical observation. Perhaps most interestingly, entirely new capabilities can emerge at certain scale thresholds that weren't explicitly programmed.

Pre-training
The initial training phase where AI models learn patterns from vast amounts of text data, developing a foundational understanding of language and knowledge.

Fine-tuning
Additional training after pre-training where models learn to follow instructions, provide helpful responses, and avoid generating harmful content.

Context window
The amount of information an AI can consider at one time, including the conversation history and any documents you've shared. Has a maximum limit that varies by model.

Hallucination
A type of error when AI confidently states something that sounds plausible, but is actually incorrect.

Knowledge cutoff date
The point after which an AI model has no built-in knowledge of the world, based on when it was trained.

Reasoning or thinking models
Types of AI models specifically designed to think step-by-step through complex problems, showing improved capabilities for tasks requiring logical reasoning.

Temperature
A setting that controls how random an AI's responses are. "Higher" temperature produces more varied and creative outputs (think boiling water bubbling), while "lower" temperature produces more predictable and focused responses (think ice crystals).

Retrieval augmented generation (RAG)
A technique that connects AI models to external knowledge sources to improve accuracy and reduce hallucinations.

Bias
Systematic patterns in AI outputs that unfairly favor or disadvantage certain groups or perspectives, often reflecting patterns in training data.

Prompt Engineering Concepts

Prompt
The input given to an AI model, including instructions and any documents shared.

Prompt engineering
The practice of designing effective prompts for AI systems to produce desired outputs. Combines clear communication with AI-specific techniques.

Chain-of-thought prompting
Encouraging an AI to work through a problem step by step, breaking down complex tasks into smaller steps that help the AI follow your thinking and deliver better results.

Few-shot learning (n-shot prompting)
Teaching AI by showing examples of the desired input-output pattern. The "N" refers to the number of examples provided. Helps the model understand what you want without lengthy explanations.

Role or persona definition
Specifying a particular character, expertise level, or communication style for the AI to adopt when responding. Can range from general roles ("speak as a UX design expert") to specific personas ("explain this like Richard Feynman would").

Output constraints / output formatting
Clearly specifying within your prompt the desired format, length, structure, or other characteristics of the AI's response to ensure you get exactly what you need.

Think-first approach
Explicitly asking the AI to work through its reasoning process before providing a final answer, which can lead to more thorough and well-considered responses.

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The AI Fluency Framework

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Four interconnected competencies necessary to ensure our interactions with AI are effective, efficient, ethical and safe.

The 4D Framework

Delegation	Setting goals and deciding whether, when and how to engage with Al.
Description	Effectively describing goals to prompt useful Al behaviors and outputs.
Discernment	Accurately assessing the usefulness of Al outputs and behaviors.
Diligence	Taking responsibility for what we do with AI and how we do it.

Key Concepts

Three Modes of AI interaction:

- **Automation**: AI executes specific tasks based on human instructions
- **Augmentation**: Humans and AI collaborate as thinking partners
- **Agency**: Humans configure AI to independently perform future tasks on their behalf

AI Fluency means interacting with AI systems in ways that are effective, efficient, ethical and safe.

The Delegation, Description, Discernment, and Diligence Competencies are interconnected collections of skills, knowledge, insights, values

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The 4Ds:
Delegation

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Delegation is making thoughtful decisions about what work is appropriate for you to do, for AI to do, or for you and AI to do together, and how to distribute those tasks.

What am I trying to do?

Problem Awareness

Clearly understanding your goals and the nature of the work before involving AI.

What do you do well?

Platform Awareness

Understanding the capabilities and limitations of different AI systems.

Ok, here's the plan

Task Delegation

Thoughtfully distributing work between humans and AI to leverage the strengths of each.

Effective delegation requires both domain expertise and an understanding of AI capabilities.

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The 4Ds:
Description

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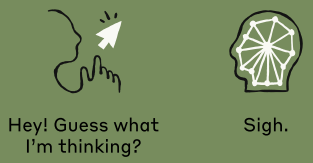
Description is the ability to communicate with AI in ways that create a productive collaborative environment.

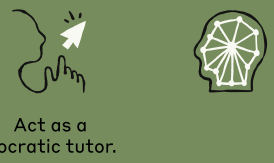


Product Description

Defining what you want in terms of outputs, format, audience, and style.

Process Description

Defining how the AI approaches your request, such as providing step by step instructions for the AI to follow.



Performance Description

Defining the AI system's behavior during your collaboration, such as whether it should be concise or detailed, challenging or supportive.

Clear communication with AI systems up front saves time and leads to better results.

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The 4Ds:
Discernment

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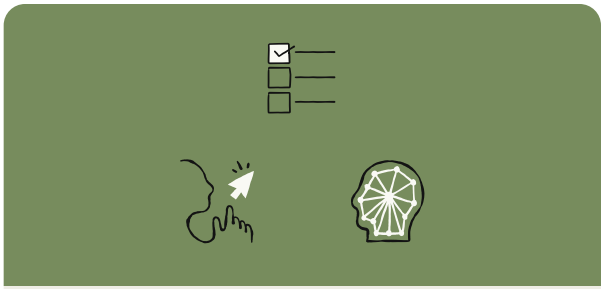
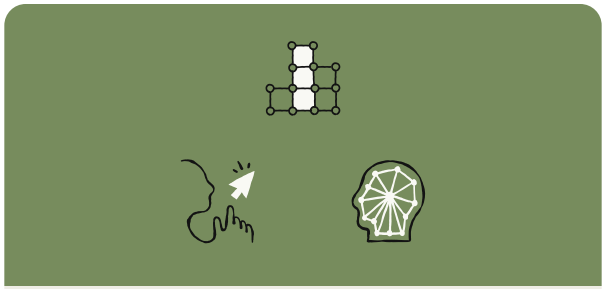
Discernment is the ability to thoughtfully and critically evaluate what AI produces, how it produces it, and how it behaves.



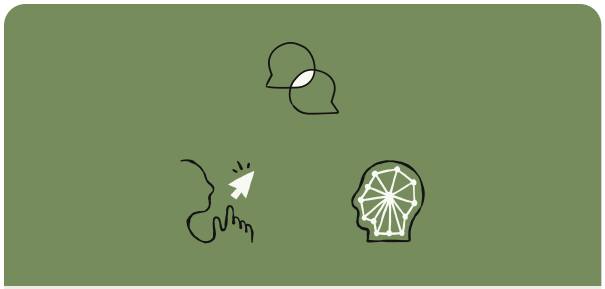
Product Discernment

Evaluating the quality of what AI produces (accuracy, appropriateness, coherence, relevance).



Process Discernment

Evaluating how the AI arrived at its output, looking for logical errors, lapses in attention, or inappropriate reasoning steps.



Performance Discernment

Evaluating how the AI behaves during your interaction, considering whether its communication style is effective for your needs.

Discernment works hand-in-hand with Description in a continuous feedback loop.

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The 4Ds:
Diligence

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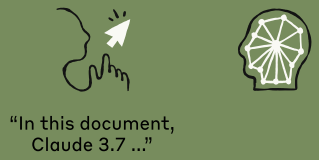
Diligence is taking responsibility for what we do with AI and how we do it.

Creation Diligence

Being thoughtful about which AI systems you use and how you interact with them.



Transparency Diligence

Being honest about AI's role in your work with everyone who needs to know.

Deployment Diligence

Taking responsibility for verifying and vouching for the outputs you use or share.

Different contexts (personal, academic, professional) may have different expectations for disclosure and verification.

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